# Practical No.1

**Objective:** Write a Program to implement the Linear and Binary Search Algorithm.

**Pseudo Code :**

Code:

```
#include<iostream>
using namespace std;
//Lineary search
int L_search(int arr[], int n, int x, int i) {
   if(i>n) {
      return -1;
   }
   else{
      if(arr[i]==x){
         return i;

      }else{
         return L_search(arr, n,x,i+1);
      }
   }
}
//Binary Search
int B_search(int arr[],int x,int low ,int high){
   if(low>high){
      return -1;
   }
   else{
      int mid=(low+high)/2;
      if(arr[mid]==x){
         return mid;
      }
```

```
        else if(arr[mid]>x){
            return B_search(arr,x,low,mid-1);
        }
        else{
            return B_search(arr,x,mid+1,high);
        }
    }
}
int main(){
    int arr[]={1,2,3,4,5,6};
    cout<<L_search(arr,6,5,0)<<endl;
    cout<<B_search(arr,5,0,5)<<endl;
}
```

Output:

